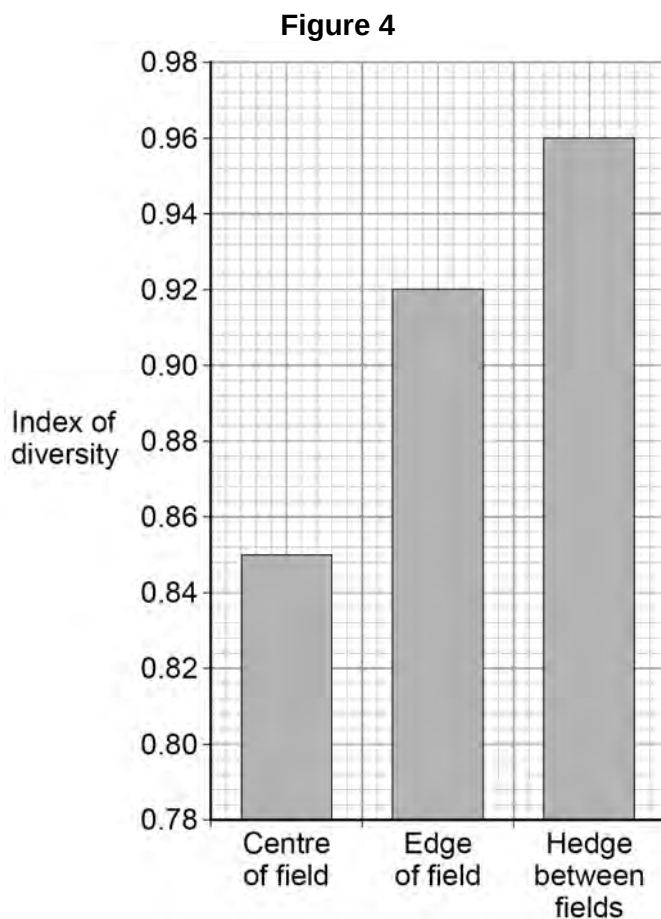


0 3

A group of students investigated biodiversity of different areas of farmland. They collected data in each of these habitats:

- the centre of a field
- the edge of a field
- a hedge between fields.

Their results are shown in **Figure 4**.

**0 3 . 1**

What data would the students need to collect to calculate their index of diversity in each habitat?

Do **not** include apparatus used for species sampling in your answer.

[1 mark]

0 3 . 2

Give **two** ways the students would have ensured their index of diversity was representative of each habitat.

[2 marks]

1 _____

2 _____

0 3 . 3

Modern farming techniques have led to larger fields and the removal of hedges between fields.

Use **Figure 4** to suggest why biodiversity decreases when farmers use larger fields.

[1 mark]

Question 3 continues on the next page

0 3 . 4

Farmers are now being encouraged to replant hedges on their land.

Suggest and explain **one** advantage and **one** disadvantage to a farmer of replanting hedges on her farmland.

[2 marks]

Advantage

Disadvantage

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

Turn over ►
